Collective Communication in AI Infra

HJ

November 3, 2026

1 Collective Communication

Collective communication is defined as communication that involves a group of processes, these concept used widely in AI Infra:

- Broadcast(1:N): 1 member send copies to all group members
- Scatter(1:N): 1 member scatter data to all.
- Gather(1:N): 1 member gather data from all.
- AllGather(N:N): Each member gather data from all, everyone should have the complete and same dataset.
- All2All(N:N): Change the datalayout by dispatch+combine, here the example did Matrix Transpose.

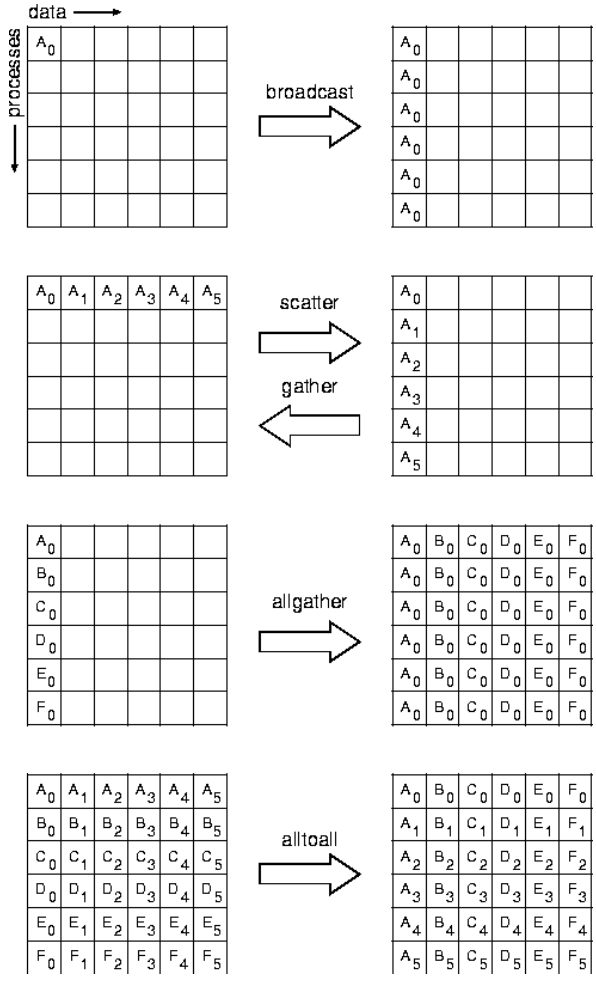


Figure 1: Collective communication patterns for six processes, from the MPI-1.1 standard [1].

2 About Reduce

TODO: Should read slides about ReduceScatter, Allgather in LLM, and AllReduce in MPI

References

[1] MPI Forum, *MPI: A Message-Passing Interface Standard*, Version 1.1, Section 4.1, "Introduction and Overview."